

P- 8

PRE-BOARD 1 (2024-25)

SUBJECT: MATHEMATICS-STANDARD (041)

CLASS-X

TIME :3Hours

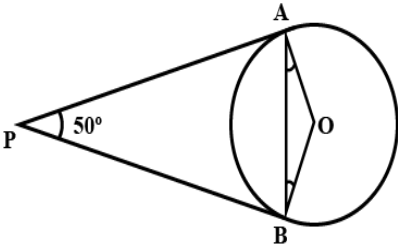
MAX. MARKS-80

General instructions:

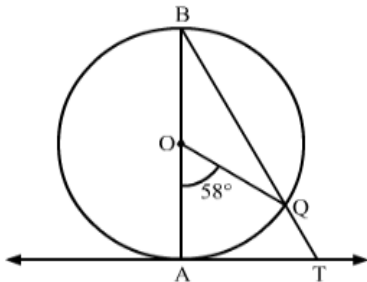
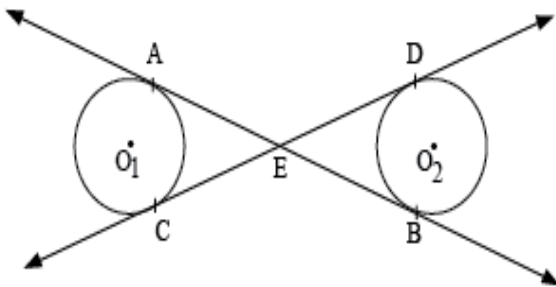
Read the following instructions carefully and follow them:

- (i) This question paper contains 38 questions
- (ii) This question paper divided into five section- A, B, C, D and E
- (iii) In section A, question no 1 to 18 are multiple choice question (MCQs) and question number 19 and 20 are Assertion – Reason based question of 1 mark each.
- (iv) In section B, question no 21 to 25 are very short answer (VSA) Type question carrying 2 marks each.
- (v) In Section C, question no 26 to 31 are short answer (SA) type questions carrying 3 marks each.
- (vi) In Section D, Question no. 32 to 35 are long answer (LA) Type questions carrying 5 marks each.
- (vii) In Section E Question no 36 to 38 are case study-based questions carrying 4 marks each. Internal choice is provided in 2 marks questions in each case study.
- (viii) There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 question in Section C, 2 question in section D and 3 questions in section E.
- (ix) Use of calculators is not allowed.
- (x) Draw neat diagram wherever required. Take $\pi = \frac{22}{7}$ wherever required, if not stated.

	SECTION-A	
1	$\triangle ABC$ and $\triangle DEF$, $\frac{AB}{DE} = \frac{BC}{FD}$ Which of the following makes the two-triangle similar (a) $\angle A = \angle D$ (b) $\angle B = \angle D$ (c) $\angle B = \angle E$ (d) $\angle A = \angle F$	1
2	If $\operatorname{cosec} \theta + \cot \theta = x$, value of $\operatorname{cosec} \theta - \cot \theta$ is	1

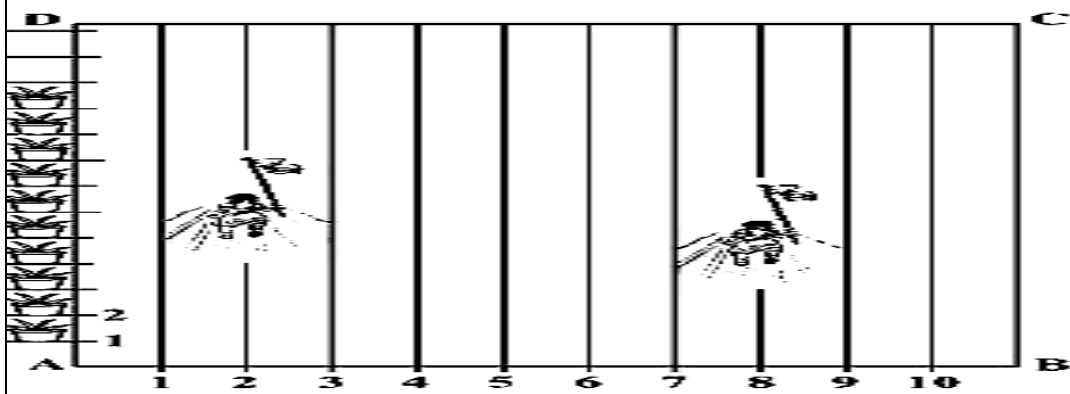
	(a) x	(b) $\frac{2}{x}$	(c) $\frac{1}{x}$	(d) 1	
3	$10\sin^2\theta + 10\cos^2\theta = ?$ (a) 1 (b) 5 (c) 10 (d) $\frac{1}{10}$				1
4	A pole 6m high casts a shadow $2\sqrt{3}$ m long on the ground then, the Sun's elevation is (a) 60° (b) 45° (c) 30° (d) 90°				1
5	The HCF and LCM of two number are 9 and 360 respectively. If one number is 45 what is the other number. (a) 50 (b) 75 (c) 73 (d) 72				1
6	The number of polynomials having zeroes as 3 and 5 is (a) 1 (b) 2 (c) 3 (d) More than 3				1
7	The distance of the point P(-6,8) from the origin is (a) 8 (b) 9 (c) 10 (d) 11				1
8	The pair of equations $x + 2y + 5 = 0$ and $-3x - 6y + 1 = 0$ have (a) Unique solution (b) Two solution (c) Infinitely many solution (d) No solution				1
9	$\triangle ABC \sim \triangle PQR$. If AM and PN are altitudes of $\triangle ABC$ and $\triangle PQR$ respectively and $AB^2 : PQ^2 = 4 : 9$ Then AM:PN is (a) 3:2 (b) 16:81 (c) 4:9 (d) 2:3				1
10	In the given figure, If PA and PB are tangents to the circle with centre O such that $\angle PAB = 50^\circ$, then $\angle OAB$ is equal to : (a) 25° (b) 30° (c) 40° (d) 50°				1
					
11	The ratio of heights of two circular cylinder of same volume is 1:3, the ratio of the radii of their base is. (a) 3:1 (b) 2:1 (c) $\sqrt{3} : 1$ (d) $\sqrt{2} : 1$				1
12	If the perimeter and the area of a circle are numerically equal, then the radius of the circle is (a) 2 units (b) 3 units (c) 5 units (d) 2 units				1
13	A card is drawn at random from a well shuffled pack of 52 playing cards. Probability of getting neither a red card nor a queen is				1

	(a) $\frac{1}{13}$	(b) $\frac{2}{26}$	(c) $\frac{6}{13}$	(d) $\frac{4}{52}$	
14	Total number of factors of a prime number is (a) 0 (b) 1 (c) 2 (d) 4				1
15	Mode+2mean=? (a)Probability (b) 3Median (c) 3Mode (d) 3Mean				1
16	Which of the following is a real number? (a) $\sqrt{-1}$ (b) $\frac{3}{0}$ (c) $\sqrt{5}$ (d) None of these				1
17	If the quadratic equation $ax^2 + bx + c = 0$ has two real and equal roots, then “c” is equal to (a) $\frac{-b}{2a}$ (b) $\frac{b}{2a}$ (c) $\frac{-b^2}{4a}$ (d) $\frac{b^2}{4a}$				1
18	Given a circle with radius 6cm and angle of sector is 60° then area of a sector of a circle is (a) $\frac{136}{7} \text{ cm}^2$ (b) $\frac{131}{7} \text{ cm}^2$ (c) $\frac{132}{7} \text{ cm}^2$ (d) $\frac{150}{7} \text{ cm}^2$				1
	Question number 19 and 20 are Assertion and Reason based carrying 1 mark each. Two statements are given, one labelled as Assertion (A) and the other is labelled as Reason (R). Select the correct answer to these question from the codes (a)(b)(c) and (d) as given below. A) Both assertion (A) and Reason (R) are true and Reason is the correct explanation of the Assertion(A) B) Both Assertion(A) And Reason (R) are true, but Reason(R) is not the correct explanation of the assertion(A) C) Assertion (A) is true but Reason (R) is false. D) Assertion (a) is false, but Reason (R) is true				1
19	Assertion (A) 34.12345 is a terminating decimal fraction. Reason (R) Denominator of 34.12345 when expressed in the form p/q, $q \neq 0$, is the form of $2^m \times 2^n$, where m and n are non-negative integer				1
20	Assertion (A): If the distance between the point (4, p) and (1,0) is 5, then the value of p is 4. Reason(R) The point which divides the line segment joining the point (7, -6) and (3,4) in the ratio 1:2 internally lies in the fourth quadrant.				1
	SECTION.B				

21	The length of a tangent from a point A at distance 5cm from the centre of the circle is 4cm. Find the radius of the circle.	2
22	Prove that $\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} = \tan \theta + \cot \theta$	2
23	The diagonals of a quadrilateral ABCD intersect each other at the point O such that $\frac{AO}{BO} = \frac{CO}{DO}$. Show that ABCD is a trapezium	2
(A)		
(B)	OR In an equilateral triangle, the length of the median is $\sqrt{3}$ cm, then find the length of the side of 2equilateral triangle.	
24	If the system of equations $6x + 2y = 3$ and $kx + y = 2$ has a unique solution, find the value of k.	2
25	In given figure ,AB is the diameter of a circle with centre O and AT is a tangent . If $\angle AOQ = 58^\circ$,Find $\angle ATQ$	2
(A)	 OR	
(B)	In the given figure common tangent AB and CD to the two circles with centre O_1 And O_2 intersect at E . Prove that AB = CD	
		
SECTION-C		
26	4 chairs and 3 tables cost Rs 2100 and 5 chairs and 2 tables	3

	costs Rs 1750. Find the cost of one chair and one table separately. OR The sum of the numerator and denominator of a fraction is 12. If 1 is added to both the numerator and denominator the fraction becomes $\frac{3}{4}$. Find the fraction.													
27	Find the quadratic polynomials whose zeroes are $3 + \sqrt{2}$ and $3 - \sqrt{2}$	3												
28	Prove that $2-3\sqrt{5}$ is an irrational number.	3												
29	Three distinct coins are tossed together. Find the probability of getting; (i) At least 2 head (ii) At most 2 heads	3												
30	Prove that length of the tangents drawn from an external point to a circle are equal	3												
31	Prove that: $\frac{\operatorname{cosec}^2 \theta}{\operatorname{cosec} \theta - 1} - \frac{\operatorname{cosec}^2 \theta}{\operatorname{cosec} \theta + 1} = 2 \sec^2 \theta$ OR Prove that: $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = \tan^2 \theta + \cot^2 \theta + 7$.	3												
	SECTION-D													
32 (A)	150 Spherical marbles each with diameter 1.4cm, are dropped in a cylindrical vessel of diameter 7cm containing some water, which are completely immersed in water. Find the rise in the level of water in the vessel. OR (B) A vessel is in the form of a hemispherical bowl surmounted by hallow cylinder of same diameter. The diameter of the hemispherical bowl is 14 cm and total height of the vessel is 13 cm. Find the total surface area of the vessel. Use $\pi=\frac{22}{7}$	5												
33	The median of the distribution given below is 14.4. Find the values of x and y . If the sum of frequency is 20. <table><tr><th>Class interval</th><th>Frequency</th></tr><tr><td>0 – 6</td><td>4</td></tr><tr><td>6 – 12</td><td>x</td></tr><tr><td>12 – 18</td><td>5</td></tr><tr><td>18 – 24</td><td>y</td></tr><tr><td>24 – 30</td><td>1</td></tr></table>	Class interval	Frequency	0 – 6	4	6 – 12	x	12 – 18	5	18 – 24	y	24 – 30	1	5
Class interval	Frequency													
0 – 6	4													
6 – 12	x													
12 – 18	5													
18 – 24	y													
24 – 30	1													

34	A truck covers a distance of 150 km at certain average speed and then covers another 200 km at an average speed which is 20km per hour more than the first speed. If the truck covers the total distance in 5 hours, find the first speed of the truck. OR Two water taps together can fill a tank in 6hours. The tap of larger diameter takes 9 hours less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank.	5
35	A girls of height 90cm is walking away from the base of a lamp post at a speed 1.2m/s. If the lamp is 3.6 m above the ground , find the length of her shadow after 4 seconds.	5
	SECTION-E	
36	India is competitive manufacturing location due to low cost of manpower and strong technical and engineering capabilities contributing to higher quality production runs. The production of TV sets in a factory increases uniformly by a fixed number every year. It produced 16000 sets in 6th years and 22600in 9th year. 	
(i)	Find the production during first year.	2
(ii)	Find the production during first three years. OR In which year, the production is Rs29200.	2
37	Annual sport day activities in your School, lines have been drawn with chalk power at a distance of 1m each, in a rectangular shaped ground ABCD,100 flowerpots have been placed at a distance of 1m from the each other along AD, as shown in the given figure. Riya run $\frac{1}{4}$th the distance AD on the 2nd line and posts a green (G) Flag. Fateha runs $\frac{1}{5}$th distance AD on eighth line and posts a red (R) Flag.	



(i) Find the coordinate of green flag. 2

(ii) Find the coordinate of red flag. 2

OR

(iii) What is distance between both Flags? 2

38 A toll plaza an electronic toll collection system has been installed. Fast tag can be used to pay the fare. The tag can be pasted on the windscreen of a car. At the toll plaza a tag scanned reads the information on the tag of the vehicle and debits the desired toll amount from a linked bank amount. For the tag scanner to function properly the speed of a car needs to be less than 30km per hours. A car with a tag installed at a height of 1.5m from the ground enters the scanner zone



(i)	The scanner gets activated when the car tag is at distance of 5m from it. Give one trigonometric ratio for the angle between the horizontal and the line between the car tag and the scanner?	2
(ii)	The scanner read the complete information on the cars tag while the angle between chip and scanner changes from 30° to 60° due to car movement. What is the distance moved by the car.	2
	OR	
(iii)	Vehicle with a tag pasted at a height of 2m from the ground stops in the scanner zone. The scanner reads the data at an angle of 45° . What is distance between the tag and the scanner.	2

*****ALL THE BEST*****